

Hybrid Federated Learning with TabTransformer and FedMADE-GSA for IoT Intrusion Detection

Trong-Thua Huynh, Hy Phan Thanh*, Nguyen Nguyen Thi Bich, Gioi Ho Le, Tri Nguyen Dinh Quoc
Department of Information Technology 2, Posts and Telecommunications Institute of Technology
Ho Chi Minh City, Vietnam

Email: {thuaht, hypt, nguyennntb}@ptit.edu.vn; {n22dcat019, n22dcat059}@student.ptithcm.edu.vn

Abstract—Federated Learning (FL) has emerged as a revolutionary paradigm for privacy-preserving machine learning across distributed Internet of Things (IoT) environments. However, contemporary federated learning-based intrusion detection systems encounter challenges such as non-IID data distributions, class imbalance, limited communication resources, and the progression of cyber threats. This paper introduces a novel hybrid federated learning framework that integrates three critical components: TabTransformer for optimal tabular data representation, FedMADE Weighting for adaptive client aggregation employing Class Probability Matrices (CPM) and client clustering through DBSCAN via a systematic optimization process, and Gradient Similarity Aggregation (GSA) for effective outlier filtration. The suggested technique is evaluated using the comprehensive CICIoT2023 dataset, comprising over 46 million samples and 34 unique attack types. Experimental results demonstrate outstanding performance, achieving 99.41% accuracy in assessments. This research introduces a scalable, efficient, and resilient federated learning approach tailored for actual IoT intrusion detection applications.

Keywords—Federated Learning, Internet of Things, Intrusion Detection, FedMADE Weighting, Non-IID Data, CICIoT2023 Dataset, TabTransformer, GSA.

I. INTRODUCTION

The Internet of Things (IoT) is growing quickly, and it is collecting a lot of different types of data from areas like healthcare, manufacturing, and transportation. This encourages new ideas, but it also makes security risks higher, which makes IoT ecosystems great places for cyberattacks. Privacy issues, communication constraints, and the complexity of dispersed networks all make traditional centralized intrusion detection systems (IDS) less useful [1]. Federated Learning (FL) resolves these challenges by facilitating collaborative training while preserving raw data confidentiality [2]. Nonetheless, FL-based IDS encounter significant obstacles: non-IID and imbalanced data, frequent distribution shifts, and

the limitations of edge devices with constrained resources [3], [4]. To solve these problems, we provide a hybrid FL framework that is built for IoT intrusion detection. The approach uses Tab-Transformer to find complicated connections between features in tabular cybersecurity data [5]. An innovative FedMADE Using DBSCAN clustering [6], the weighting strategy constantly changes how much each client contributes. Gradient Similarity Aggregation (GSA) makes the system more resilient by filtering out strange updates using cosine similarity [7]. Also, adaptive client grouping makes it easier to scale when there is a lot of different data. A lot of testing on the CICIoT2023 dataset, which has 46 million traffic samples and 34 attack types, demonstrates that the suggested system is quite accurate, converges quickly, and is resistant to modifications that are meant to hurt it. Our efforts combine transformer-based modeling with adaptive aggregation to make FL-based IDS safer and more scalable for real-world IoT settings.

II. RELATED WORK

A. CICIoT2023 and Deep Learning in IDS

In IoT environments, intrusion detection systems (IDS) must be evaluated using realistic datasets. The IDS evaluation benchmark CICIoT2023, published by Pinto Neto et al. [8], contains approximately 46 million data from 33 attack types. This resource allowed Gheni and Al-Yaseen [9] to create a two-stage clustering procedure that improved training and performance ($F1 = 97.2\%$). Many deep-learning methods use CICIoT2023. The work in [10] employed an LSTM architecture to discover traffic patterns over time with 98.5% accuracy and above 97% precision. Hafezian et al. [11] chose a multi-class baseline of 96.8% for decision trees, random forests, and SVMs. Zhang et al. [12] proposed a Transformer-based detector that outperformed LSTM and CNN baselines in class imbalance. The F1-score was 98.9%. We also

*Corresponding author: hypt@ptit.edu.vn

considered lightweight designs: On resource-constrained IoT devices, Thereza and Ramli [13] showed that rigorous feature selection and shallow models can achieve over 95% accuracy. Recently, Gulzar and Mustafa [14] demonstrated DeepCLG, a hybrid CNN-graph architecture with over 99% detection accuracy and strong resistance to complex industrial IoT assaults.

B. Federated Learning for IoT IDS

Initial research, including Mothukuri et al. [15], initiated the application of federated learning (FL) in anomaly detection, demonstrating significant accuracy while maintaining anonymity. Xie et al. [16] enhanced scalability in IoT by asynchronous parameter sharing, whereas Zhang et al. [17] emphasized issues such non-IID data and communication overhead, proposing hierarchical FL and personalization. Chen et al. [18] further diminished bandwidth expenses using model compression while maintaining performance integrity. Subsequent studies developed adaptable and self-enhancing FL-based IDS. Bensaid et al. [19] put forward SA-FLIDS, a secure and verified FL-based IDS for smart healthcare, whereas Albanbay et al. [20] tested how well FL-based IDS works and how well it can grow in IoT networks.

Soon after, more advanced architectures came along: Works in [21], [22] demonstrated TabTransformer's efficacy in modeling feature interactions, whereas the work [23] used CGAN with TabTransformer to address data imbalance. Optimizations have also looked into hybrid and nature-based methods. Fatema et al. [24] developed the Federated XAI IDS, which merges FL with SHAP to improve explainability while protecting privacy. Li et al. [25] put forward a server-edge hybrid FL that works well in transportation IoT. The work [26] utilized the Chimp algorithm for feature optimization, whereas Anwar et al. [27] employed FL-based LSTM to address temporal dynamics in sensor networks. Zhang et al. [7] presented NIDS-FGPA from an aggregation perspective, utilizing Gradient Similarity Aggregation (GSA) to counteract adversarial updates.

Lastly, personalization and diversity have become quite important. The work [28] introduced StatAvg to address data heterogeneity in federated learning-based intrusion detection systems, while Singh et al. [29] developed a federated learning framework for IoT intrusion detection utilizing TabTransformer and nature-inspired hyperparameter optimization.

III. METHODOLOGY

FL-based IDS for IoT face challenges such as non-IID data, imbalance, communication overhead, adversarial vulnerability, and ineffective aggregation. We propose a hybrid FL framework combining TabTransformer with

FedMADE weighting and GSA filtering, alongside hybrid server-client coordination. Local TabTransformers are trained at clients, while the server applies adaptive weighting and anomaly filtering to construct a more robust global model. Figure 1 illustrates the proposed

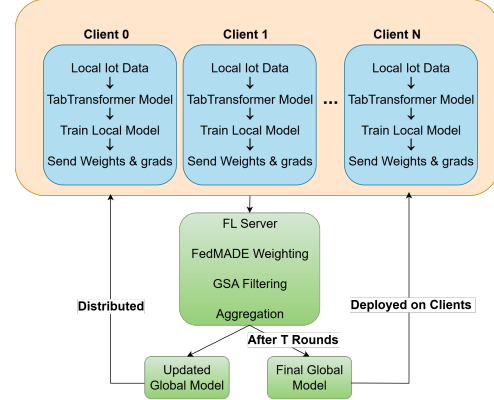


Fig. 1. Proposed hybrid FL architecture combining TabTransformer, FedMADE, and GSA for IoT intrusion detection.

architecture. Each IoT client retains local data and trains a TabTransformer model independently. The learned parameters are then transmitted to the central FL server, where FedMADE weighting, GSA filtering, and aggregation are applied to construct an improved global model.

A. FedMADE Weighting Strategy

FedMADE assigns dynamic client weights during aggregation, overcoming the limits of FedAvg by optimizing weights through Class Probability Matrices rather than simple accuracy or gradient norms. [6]. Step 1 – Class Probability Matrix (CPM): For each client model ω_i , the server computes a CPM that reflects its classification behavior across all classes:

$$P_{\omega_i} = [\bar{P}_1, \bar{P}_2, \dots, \bar{P}_{N_C}], \quad \bar{P}_C = \frac{1}{n_c} \sum_{j=1}^{n_c} p(x_{c,j}) \quad (1)$$

where n_c is the number of validation samples in class c , and $p(x_{c,j})$ is the probability output vector for sample $x_{c,j}$.

Step 2 – Client Clustering: Using DBSCAN [6], clients with similar CPMs are grouped into clusters C_k . Step 3 – Weight Optimization: The aggregation weights for clusters are obtained by solving the following optimization problem:

$$\min \left\| \sum_{k=1}^K \alpha_k P_{C_k} - I \right\|_2, \quad \alpha_k \geq 0 \quad (2)$$

where I is the identity matrix representing ideal classification (100% correct per class). Clusters whose CPMs

are closer to the ideal receive higher weights. The optimization is performed using Adam with learning rate 0.001 and maximum 100 iterations, parameters tuned for stable convergence.

Step 4 – Client-Level Weights: For each client $i \in C_k$, its weight is derived from the cluster weight:

$$\beta_i = \frac{\alpha_k}{|C_k|} \quad (3)$$

All client weights are normalized so that $\sum_i \beta_i = 1$.

Step 5 – Global Aggregation: The new global model is updated as:

$$\omega^{(r+1)} = \sum_i \beta_i \omega_i^{(r)} \quad (4)$$

This formulation ensures that clients demonstrating consistently strong classification performance across all categories, including minority attacks, are granted greater weight in aggregation, while unstable or compromised clients are inherently diminished due to their CPM divergence.

We carefully adjusted numerous FedMADE parameters in our implementation to get the best performance. We specifically chose DBSCAN with $\epsilon = 0.3$ and $\text{minPts} = 3$, which gave us reliable clustering of different IoT clients based on validation tests. Adam got rapid and steady convergence for weight optimization. We also improved the normalization factors so that no one cluster would take over the aggregation. These parameter selections yielded enhanced accuracy and robustness relative to prior configurations.

B. Gradient Similarity Aggregation (GSA)

The cosine similarity [30] is defined as:

$$\text{sim}(\vec{g}_i, \vec{g}_j) = \frac{\vec{g}_i \cdot \vec{g}_j}{\|\vec{g}_i\| \|\vec{g}_j\|} \quad (5)$$

We employ Gradient Similarity Aggregation (GSA) to safeguard against unreliable or untrustworthy clients. This method calculates the cosine similarity between the gradient of each client and the mean gradient of all clients. Updates with a similarity below a specified threshold (0.7) are assigned diminished significance or excluded entirely. This technique identifies anomalous updates regardless of the gradient’s magnitude, enhancing the system’s resilience against malicious contributions.

C. TabTransformer

In our framework, TabTransformer processes IoT tabular data consisting of categorical and continuous features. Categorical features [31] are embedded into vectors of dimension $d_{\text{model}} = 128$, forming:

$$E = [e_1, e_2, \dots, e_k], \quad e_i \in \mathbb{R}^{d_{\text{model}}} \quad (6)$$

These embeddings are sequentially processed through 6 Transformer encoder layers (depth = 6).

We integrate the output from the categorical branch with normalized continuous data and process it through a multilayer perceptron classifier (dropout = 0.3) to predict the class of attack. Employing class-weighted cross-entropy, each client conducts local learning utilizing Adam (learning rate = 0.001, batch size = 256, 30 epochs). Upon completion of training, clients transmit the model parameters, validation accuracy, and gradient norms to the server. FedMADE does global aggregation, while GSA refines the results to ensure the updates are robust and dependable.

D. Problem Analysis and Proposal

Prior research applied federated learning (FL) to anomaly detection, mostly via FedAvg, which assumes IID data and lacks robustness [15], [16]. GSA was introduced by Zhang et al. [7], but without integration into advanced designs. TabTransformer has proven effective for tabular data in centralized settings [21], [22], while Abd Elaziz et al. [24] explored it in FL with a focus on hyperparameter tuning. Adaptive weighting methods [28] improved aggregation but often ignored accuracy–convergence trade-offs.

In short, existing work faces four key limitations: (i) unrealistic IID assumptions, (ii) weak integration of GSA with adaptive weighting, (iii) limited use of TabTransformer in FL, and (iv) inadequate weighting strategies. To address these, our framework combines TabTransformer for feature modeling, FedMADE Weighting with CPMs and DBSCAN for balanced client contributions, and GSA for filtering malicious updates ensuring robustness, fairness, and efficiency in real-world IoT scenarios.

IV. EXPERIMENTAL RESULTS

We evaluated our model on the CICIOT2023 dataset (46M samples, 34 attack types), splitting data at the flow level with stratified sampling (70% training, 20% validation, 10% testing) to ensure non-overlap and realistic non-IID distributions. Heterogeneity and imbalance were further introduced by partitioning into two clients via KMeans. The proposed framework achieved 99.41% accuracy, surpassing prior methods in robustness, efficiency, and resilience to poisoned updates. By integrating TabTransformer for feature learning, FedMADE for adaptive client weighting, and GSA for anomaly filtering, our approach enhances both performance and reliability in federated IoT intrusion detection.

Table I shows how the best federated intrusion detection algorithms compare on the CICIOT2023 dataset. As shown, our proposed framework combining TabTransformer, FedMADE Weighting, and Gradient Similarity

Algorithm 1: Improved Aggregation Algorithm:
Hybrid FL with FedMADE + GSA

Input: Total rounds R , client sampling rate γ , distributed training dataset, validation dataset for CPM

Output: Final global model $\omega^{(R)}$

for each round $r = 1, \dots, R$ **do**

Select client subset U_r ;

Broadcast global model $\omega^{(r)}$ to all clients in U_r ;

for each client $i \in U_r$ **do**

Preprocess local data with SMOTE and RobustScaler;

Train local TabTransformer $\rightarrow \omega_i^{(r)}$;

Compute class probability matrix (CPM) $P_{\omega_i}^{(r)}$;

Cluster clients using DBSCAN based on CPMs $\{P_{\omega_i}^{(r)}\} \rightarrow \{C_k\}$;

for each cluster C_k **do**

Compute cluster CPM:

$P_{C_k} = \frac{1}{|C_k|} \sum_{\omega_i \in C_k} P_{\omega_i}^{(r)}$;

Optimize cluster weights α_k ;

Assign client weights: ;

$\beta_i = \frac{\alpha_k}{|C_k|}, \forall i \in C_k$;

Normalize such that $\sum_i \beta_i = 1$;

Aggregate global model::

$\hat{\omega}^{(r+1)} = \sum_{i \in U_r} \beta_i \omega_i^{(r)}$;

/* (C) Refinement with GSA */

Apply GSA refinement: ;

$\omega^{(r+1)} \leftarrow \text{GSA}(\hat{\omega}^{(r+1)})$;

TABLE I
OVERALL COMPARISON WITH RECENT METHODS ON THE
CICIoT2023

Architecture	Accuracy	Data Type	Defense Capability Summary
CNN + FL [20]	94.9%	IID	Performance evaluation only, no adversarial defense
SA-FLIDS [19]	99.09%	IID	Secure FL with blockchain, resistant to malicious clients
FedXAIIDS [24]	88.2%	IID	Explainable FL, no poisoning defense
FL + Gradient Similarity [7]	99.2%	Non-IID	Partial robustness, lacks adaptive weighting
StatAvg [29]	97.6%	Non-IID	Improves normalization, no gradient filtering
Federated TabTransformer [30]	98.8%	Non-IID	TabTransformer, hyperparameter optimization, robust
TabTransformer + FedMADE + GSA	99.41%	Non-IID	Robust, efficient, resistant to poisoned updates

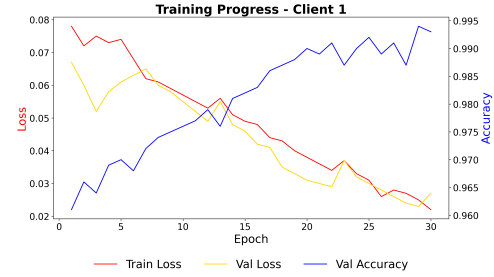


Fig. 3. Training progress of Client 1 across 30 epochs/round.

Aggregation (GSA) achieves the highest accuracy of 99.41%, while also demonstrating robustness to non-IID data distributions and strong resistance against poisoned updates. Each client trained a local TabTransformer model for 30 epochs per round, over a total of three global communication rounds.

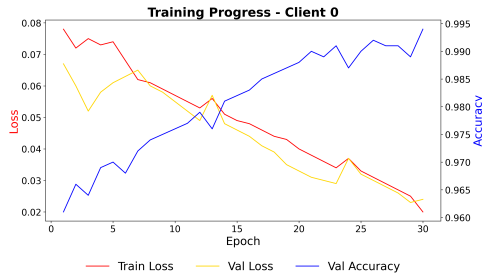


Fig. 2. Training progress of Client 0 across 30 epochs/round.

Figures 2 and 3 illustrate that despite non-IID data,

both clients converge with steadily decreasing loss, confirming the effectiveness of the proposed aggregation method across heterogeneous IoT devices. After local training, client updates were processed on the server through three steps: (1) GSA for filtering anomalous updates via cosine similarity, (2) FedMADE weighting using CPMs and DBSCAN to assess client contributions, and (3) weighted gradient aggregation to update the global model.

Figures 4 and 5 illustrate the confusion matrices of two clients using a subset of 5 representative attack types from the dataset, which are selected as representatives among the 34 attack types. The results show that the proposed model achieves consistently high classification accuracy across different clients, with most classes exceeding 96–100% correctness, demonstrating its robustness and generalization ability in federated settings.

Table II provides an overview of the proposed

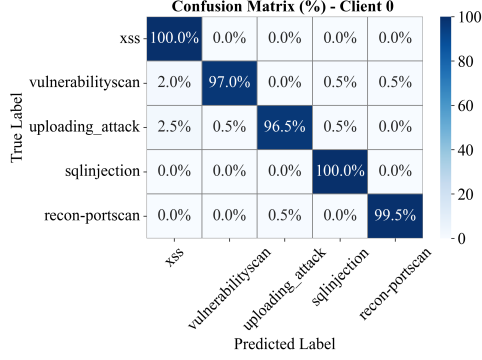


Fig. 4. Confusion matrix of Client 0 for 5 representative classes.

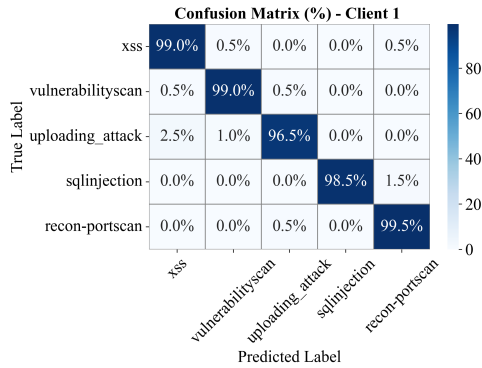


Fig. 5. Confusion matrix of Client 1 for 5 representative classes.

model's performance. The results indicate that the model achieves very high accuracy and balanced precision, recall, and F1-score. In addition, the low gradient norm (0.535) and cosine similarity values (-0.8463 to 0.9310) confirm stable training and effective collaboration between clients. Overall, the model converges reliably even under non-IID data distributions, demonstrating robustness and consistency in federated IoT intrusion detection.

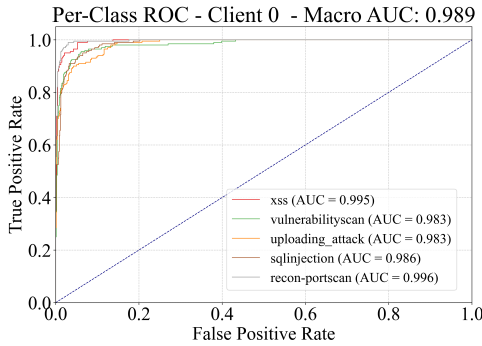


Fig. 6. ROC curves for 5 classes and Macro AUC of Client 0.

Figures 6 and 7 show the ROC curves for clients 0

TABLE II
PERFORMANCE OF THE PROPOSED MODEL

<i>Metric</i>	<i>Value</i>
Accuracy	99.41%
Precision	0.995
Recall	0.992
F1-score	99.45%
Gradient Norm	0.535
Cosine Similarity (Client 0-1)	~ -0.8463 to 0.9310
Label Distribution	Non-IID

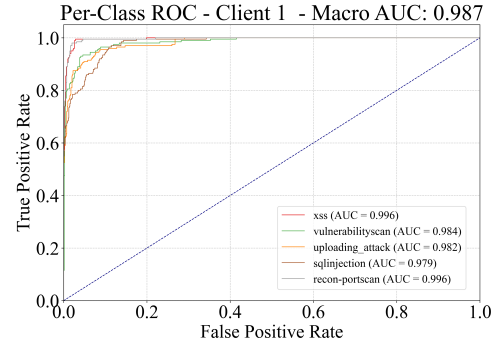


Fig. 7. ROC curves for 5 classes and Macro AUC of Client 1.

and 1 using 5 of the 34 attack types in the dataset. Both clients have very high AUC values, with macro AUCs of 0.989 and 0.987, respectively. Individual attack classes also have good detection performance, with AUC values between 0.979 and 0.996. This shows that the proposed model can reliably tell the difference between different types of attacks in federated situations.

V. CONCLUSION

In this research, we put forward this hybrid federated learning setup that mixes in three main bits: the Tab-Transformer to handle structured IoT data, FedMADE weighting for kind of dynamically adjusting how much each client contributes, using those Class Probability Matrices and DBSCAN clustering, and Gradient Similarity Aggregation to filter out any weird updates. We ran tests on the CICIOT2023 dataset in non-IID setups, and it hit 99.41% accuracy with a 99.45% F1-score, which basically shows it can deal with noisy or malicious gradients pretty well. Two critical issues that were identified in prior research are addressed by our methodology: (1) FedMADE's dual-weighting addresses the dispersion of non-IID data; (2) GSA's gradient similarity filtering mitigates the issue of erroneous or inconsistent updates.

However, there are several things that make it hard: (1) Scalability assessment is rather limited, as the basic

application necessitates a larger client base in practical situations; (2) Currently, there are no appropriate options for modifying the model to meet particular client demands. In the future, we will do more scalability assessments in distributed settings, use continuous learning to make changes in real time, and look into ways to personalize services to boost performance for certain clients.

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REFERENCES

- [1] S. Kumar, P. Sharma, S. Y. Moon, and J. H. Park, "Intelligent deep federated learning model for enhancing security in internet of things enabled edge computing environment," *Scientific Reports*, vol. 14, p. 88163, 2024.
- [2] L. Chen, X. Wang, Y. Zhang, M. Liu, and S. Sun, "An optimal federated learning-based intrusion detection for IoT environment," *Scientific Reports*, vol. 14, p. 93501, 2024.
- [3] Z. Qu, Q. Liu, and X. Yang, "Non-IID data challenge in federated learning: A survey," *IEEE Internet of Things Journal*, vol. 10, no. 4, pp. 2501–2518, 2023.
- [4] M. Mohammadi, A. Al-Fuqaha, M. Guizani, and J. Oh, "Semi-supervised deep reinforcement learning in support of IoT and smart city services," *IEEE Internet of Things Journal*, vol. 5, no. 2, pp. 624–635, 2018.
- [5] X. Huang, R. Zhang, and Y. Lee, "TabTransformer: Tabular data modeling using contextual embeddings," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 6472–6482, 2020.
- [6] S. Sun, et al., "FedMADE: Robust federated learning for intrusion detection in IoT networks using a dynamic aggregation method," unpublished, 2024.
- [7] T. Zhang, X. Liu, and J. Chen, "NIDS-FGPA: A federated learning network intrusion detection algorithm based on secure aggregation of gradient similarity models," in *Proc. IEEE Int. Conf. Communications (ICC)*, pp. 1117–1124, n.d.
- [8] E. C. Pinto Neto, et al., "CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment," *Sensors*, vol. 23, no. 13, p. 5941, 2023.
- [9] H. Q. Ghenni and W. L. Al-Yaseen, "Two-step data clustering for improved intrusion detection system using CICIoT2023 dataset," *e-Prime - Advances in Electrical Engineering, Electronics and Energy*, vol. 9, p. 100673, 2024.
- [10] A. I. Jony and A. K. B. Arnob, "A long short-term memory-based approach for detecting cyber attacks in IoT using CIC-IoT2023 dataset," *Journal of Edge Computing*, vol. 3, no. 1, pp. 28–42, 2024.
- [11] Z. Hafezian, M. Naderan, and M. Jaderyan, "A machine learning-based approach for multi-class intrusion detection and classification in IoT using CICIoT2023 dataset," in *Proc. 2024 11th Int. Symp. Telecommunications (IST)*, pp. 161–165, 2024.
- [12] Y. Zhang, X. Wang, and L. Zhang, "Multi-class intrusion detection based on Transformer for IoT networks using CIC-IoT2023 dataset," *Future Internet*, vol. 15, no. 4, pp. 112–128, 2023.
- [13] N. Thereza and K. Ramli, "Development of intrusion detection models for IoT networks utilizing CICIoT2023 dataset," in *Proc. 3rd Int. Conf. Smart Cities, Automation and Intelligent Computing Systems (ICON-SONICS)*, pp. 66–72, 2023.
- [14] Q. Gulzar and K. Mustafa, "Enhancing network security in industrial IoT environments: A DeepCLG hybrid learning model for cyberattack detection," *Int. J. Machine Learning and Cybernetics*, 2025, doi: 10.1007/s13042-025-02038-y.
- [15] V. Mothukuri, P. Khare, R. M. Parizi, S. Pouriyeh, A. Dehghan-tanha, and G. Srivastava, "Federated learning-based anomaly detection for IoT security attacks," *IEEE Internet of Things Journal*, vol. 8, no. 7, pp. 4398–4407, 2021.
- [16] H. Xie, M. Xia, P. Wu, S. Wang, and K. Huang, "Decentralized federated learning with asynchronous parameter sharing for large-scale IoT networks," *IEEE Internet of Things Journal*, vol. 11, no. 2, pp. 1734–1748, 2024.
- [17] T. Zhang, et al., "Federated learning for Internet of Things: Applications, challenges, and opportunities," *IEEE Internet of Things Journal*, vol. 9, no. 2, pp. 1111–1132, 2022.
- [18] H. Chen, S. Huang, D. Zhang, M. Xiao, M. Skoglund, and H. V. Poor, "Federated learning over wireless IoT networks with optimized communication and resources," *IEEE Transactions on Communications*, vol. 69, no. 11, pp. 7612–7627, 2021.
- [19] R. Bensaid, et al., "SA-FLIDS: Secure and authenticated federated learning-based intelligent network intrusion detection system for smart healthcare," *PeerJ Computer Science*, vol. 11, p. e2414, 2024.
- [20] N. Albanbay, Y. Tursynbek, K. Graffi, R. Uskenbayeva, Z. Kalpeyeva, Z. Abilkaiyr, and Y. Ayapov, "Federated learning-based intrusion detection in IoT networks: Performance evaluation and data scaling study," *Journal of Sensor and Actuator Networks*, vol. 14, no. 4, p. 78, 2025.
- [21] Y. She, "A robust PPO-optimized Tabular Transformer Framework for Intrusion Detection in Industrial IoT Systems," in *Proc. 42nd Int. Conf. Machine Learning (ICML)*, vol. 97, pp. 6893–6901, 2025.
- [22] B. Bazaluk, M. Hamdan, M. Ghaleb, M. S. M. Gismalla, F. S. Correa da Silva, and D. M. Batista, "Towards a transformer-based pre-trained model for IoT traffic classification," in *Proc. 2024 IEEE Network Operations and Management Symposium (NOMS)*, pp. 1–7, 2024, doi: 10.1109/NOMS59830.2024.10575448.
- [23] S. Shen, S. Yin, S. Liang, and H. Geng, "Research on abnormal traffic classification algorithm based on CGAN and TabTransformer," in *Proc. SPIE 13562, International Conference on Computer Application and Information Security (ICCAIS 2024)*, Art. no. 1356244, Apr. 25, 2025, doi: 10.1117/12.3060585.
- [24] K. Fatema, et al., "Federated XAI IDS: An explainable and safeguarding privacy approach to detect intrusion combining Federated Learning and SHAP," *Future Internet*, vol. 17, no. 6, p. 234, 2025.
- [25] H. Li, Y. Zhao, and T. Wang, "Fine-tuning federated learning-based intrusion detection systems for transportation IoT," in *Proc. 2025 IEEE Int. Conf. Internet of Things and Intelligence Systems (IoTAIS)*, vol. 3, pp. 134–140, 2025.
- [26] A. Karunamurthy, K. Vijayan, P. R. Kshirsagar, et al., "An optimal federated learning-based intrusion detection for IoT environment," *Sci. Rep.*, vol. 15, Art. no. 8696, 2025, doi: 10.1038/s41598-025-93501-8.
- [27] R. W. Anwar, M. Ahmad, and H. T. Rauf, "Federated learning with LSTM for intrusion detection in IoT-based wireless sensor networks: A multi-dataset analysis," in *Proc. 2025 IEEE Int. Conf. Artificial Intelligence and Computer Applications (ICAICA)*, vol. 1, pp. 205–213, 2025.
- [28] P. S. Bouzinis, et al., "StatAvg: Mitigating Data Heterogeneity in Federated Learning for Intrusion Detection Systems," *arXiv preprint arXiv:2405.13062*, pp. 1–12, 2024.
- [29] A. Singh, R. Kumar, P. Sharma, and M. Alazab, "Federated learning framework for IoT intrusion detection using tab transformer and nature-inspired hyperparameter optimization," *Frontiers in Big Data*, vol. 8, p. 1526480, 2025.
- [30] Z. Zhou, Y. Li, X. Ren, and S. Yang, "Towards efficient and stable K-Asynchronous Federated Learning with unbounded stale gradients on Non-IID data," *Cosine Similarity*, vol. 7, 2023.
- [31] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. 31st Int. Conf. Neural Information Processing Systems (NeurIPS)*, pp. 4–5, 2017.